

A Four-Axis Systematization of AI and Cybersecurity in the Era of Large Language Models

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Abstract—The convergence of artificial intelligence (AI) and cybersecurity has created a complex ecosystem of capabilities and threats. We systematize this space along four axes: AI-for-Cyber, Cyber-for-AI, LLM-for-Cyber, and Cyber-for-LLMs. Using a PRISMA-guided review of 422 recent works, we develop a unified taxonomy of AI-cyber interactions. AI-for-Cyber enables anomaly detection, intrusion detection, vulnerability analysis, and automated response. Cyber-for-AI highlights threats such as data poisoning, model extraction, and privacy leakage. Cyber-for-LLMs exposes vulnerabilities including prompt injection and jailbreaking, while LLM-for-Cyber introduces reasoning-driven capabilities for threat intelligence and offensive operations. We further identify compound cross-axis threats, where attackers combine techniques across axes (e.g., LLM-generated attacks exploiting AI defenses). This Systematization of Knowledge (SoK) consolidates fragmented literature, highlights systemic risks, and outlines open challenges for securing AI-enabled cyber systems.

Index Terms—Artificial Intelligence, Cybersecurity, Large Language Models, AI for CyberSecurity, Cybersecurity for AI, LLMs for Cybersecurity, CyberSecurity for LLMs

I. INTRODUCTION

Artificial Intelligence (AI) and cybersecurity have converged significantly over the past decade due to rapid technological advancements and increasing adversarial sophistication [1]. Breakthroughs in large language models (LLMs), multimodal foundation models (MFMs), and autonomous AI agents now influence core cybersecurity workflows [53], [187], [78], [201], [174], including cyber threat intelligence (CTI) enrichment [171], [11], [44] vulnerability reasoning [139], malware classification [5], security operations center (SOC) automation [43], [39], [146], and even offensive cyberspace operations [157], [175], [78], [181], [91]. The threat landscape has also evolved to target AI systems, with attacks affecting the entire AI lifecycle, including training-time data poisoning [75], inference-time adversarial perturbations [61], model extraction and theft [84], privacy leakage and membership inference [188], as well as LLM-specific vulnerabilities such as prompt injection [153], jailbreaking [153], indirect prompt manipulation, agent hijacking, and backdoor implantation [140], [174], [188]. These developments signal a fundamental shift: AI has transformed from a mere tool to a strategic asset and high-value attack surface.

AI techniques have demonstrably enhanced core cyber defense capabilities by automating the detection of intrusions,

malware, phishing, fraud, and software vulnerabilities [187], [184]. Transformer-based LLMs enable systems to reason over unstructured security data and participate in operational workflows. [78], [201].

This acceleration has created a pronounced dual reality: LLMs augment CTI pipelines and decision-making while lowering adversarial barriers, [41], [169], increasing pressure on defensive AI systems to adapt rapidly under uncertainty. Despite the growth of research in AI and cybersecurity, the literature remains fragmented across subdisciplines like adversarial machine learning (AdvML) [75], [61], LLM security [188], [174], [140], CTI automation [181], malware classification [91], secure MLOps [84], and AI governance and risk management [76], [177]. This leads to disconnected taxonomies and methodologies [84], obscuring critical systemic interdependencies. For example, AI-generated malware directly impacts ML-based detection pipelines [91]; LLM-enabled phishing reshapes email filtering and user-behavior modeling [41]; multimodal jailbreaks compromise AI-powered SOC assistants [174]; and AI governance frameworks influence both AI safety practices and cyber risk management strategies [76], [177]. Without a unified framework, research progresses limiting real-world defense effectiveness.

This fragmentation leads to the *compound interaction problem*, where AI-driven attacks, defenses, and governance mechanisms interact in complex and recursive ways. Addressing this requires a unified perspective that captures cross-axis dependencies and feedback loops. To this end, we introduce a four-axis taxonomy:

- 1) **AI-for-Cyber:** The application of AI and ML techniques to strengthen cyber defense and monitoring tasks, including intrusion detection, malware analysis, anomaly detection, and CTI enrichment [181], [91].
- 2) **Cyber-for-AI:** The use of cybersecurity principles, adversarial techniques, and defensive mechanisms to protect general AI/ML systems from integrity, confidentiality, and availability threats, such as data poisoning, adversarial examples, model extraction, and privacy leakage [84], [75].
- 3) **Cyber-for-LLMs:** Security challenges and attack vectors unique to LLMs, primarily involving linguistic and prompt-based exploits, including prompt injection,

jailbreaking, indirect manipulation, content filtration bypasses, and agent hijacking [140], [174].

- 4) **LLM-for-Cyber:** The direct operational use of LLMs in cybersecurity tasks, spanning automated CTI generation, vulnerability classification, SOC augmentation, exploit reasoning, and accelerated offensive cyber operations [78], [41].

This taxonomy systematically frames AI–cybersecurity interactions, revealing dependencies and analyzing threats that arise at their intersection. Unlike previous surveys that examine individual domains in isolation, this work offers an integrated perspective on AI–cybersecurity co-evolution.

A. Scope and Contributions

This paper addresses fragmentation in AI–cybersecurity research through three primary contributions:

1. **Unified Four-Axis Taxonomy:** We present a unified taxonomy that illustrates the dynamic interactions between AI and cybersecurity. It maps AI-driven capabilities and threats onto established frameworks like MITRE ATT&CK, D3FEND, NIST CSF, and ATLAS, bridging academic research and operational practice.

2. **PRISMA-Grounded Literature Synthesis:** We conduct a structured literature synthesis using a PRISMA-based methodology, analyzing over 422 peer-reviewed and preprint studies from 2021 to 2025, ensuring comprehensive coverage of AI, LLMs, cybersecurity, and governance.

3. **Open Research Challenges:** We identify critical challenges at the AI–cybersecurity intersection, including securing multimodal LLMs, evaluating agent-based AI systems, AI supply chain security, long-context and memory poisoning attacks, dual-use risk containment, integration with governance frameworks, and accountability in AI-enabled cyber systems.

B. Why Now? Timeliness and Necessity

Three developments highlight the urgency.

Rapid Adoption of LLMs in Cybersecurity: Enterprises, SOC, and CERTs are increasingly using LLMs for CTI summarization, alert triage, log analysis, attack surface mapping, and vulnerability prioritization, often without rigorous security evaluations, leading to new operational risks.

Accelerating Offensive AI Capabilities: Adversaries are quickly leveraging AI for malware generation, autonomous exploit development, and reconnaissance, outpacing defensive research and evaluation methodologies.

Governance Frameworks Lack Technical Grounding: Current AI governance frameworks remain high-level, offering limited technical guidance on threat models, cybersecurity integration, evaluation mechanisms, and safe deployment practices. A technically grounded systematization is essential for effective policy and standards.

Unlike prior surveys, our work introduces the concept of compound cross-axis threats and systematically analyzes how interactions between AI-driven systems create emergent vulnerabilities not captured in isolated studies.

II. METHODOLOGY

We conducted a PRISMA-based systematic review [127], targeting AI, ML, LLMs, and cybersecurity research across major academic databases, including IEEE Xplore, ACM Digital Library, SpringerLink, and ScienceDirect.

Search queries were based on four conceptual axes, adapted from established methodologies in prior Systematization of Knowledge papers. Boolean queries targeted titles, abstracts, keywords, and full text where available. Representative query patterns included:

- (“large language model” OR “LLM” OR “transformer” OR “foundation model”) AND (“security” OR “cybersecurity” OR “attack” OR “defense” OR “malware” OR “threat intelligence” OR “adversarial”)
- (“adversarial example” OR “data poisoning” OR “model extraction”) AND (“AI security” OR “ML security” OR “LLM vulnerability”)
- (“phishing” OR “APT” OR “intrusion detection”) AND (“deep learning” OR “transformer” OR “LLM”)
- (“knowledge graph” OR “cyber ontology” OR “CTI pipeline”) AND (“AI” OR “machine learning” OR “LLM”)

These strategies retrieved 2,438 documents. During screening, we excluded duplicates, non-AI cybersecurity papers, pure AI/ML theory papers without security relevance, and simulation-only cyber papers without AI components, leaving 1,584 papers. We further excluded opinion pieces, non-technical articles, and short workshop posters, reducing the candidate set to 880 papers.

A full-text eligibility review included papers that satisfied at least one of the following: (a) Proposed an AI/ML/LLM model for a cybersecurity task, (b) Studied attacks or vulnerabilities in ML or LLM systems, (c) Introduced a benchmark, taxonomy, framework, or ontology for AI–cyber interactions, (d) Presented empirical evaluations of adversarial ML, red-teaming, or LLM misuse, (e) Contributed to AI security governance, risk modeling, or assurance frameworks. Excluded were pure cryptography papers, business-focused security papers, and cyber policy papers lacking technical analysis. The final corpus comprised 422 papers.

We acknowledge that PRISMA processes may under-represent preprints with incomplete metadata and that LLM vulnerability research evolves rapidly. Nonetheless, this methodology provides a comprehensive synthesis of AI–cybersecurity research across defensive, offensive, and governance dimensions.

III. AI FOR CYBER (AI → CYBER)

Artificial Intelligence (AI) and Machine Learning (ML) are now essential to modern cybersecurity [112], [50], [?], due to the limitations of traditional rule-based defenses [14], [130]. Early systems, like signature-based intrusion detection systems (IDS), effectively addressed known threats but struggled with scalability and zero-day attacks [148]. This prompted the shift to ML-driven approaches that learn behavioral patterns and adapt to evolving threats [21], [85].

Within the surveyed literature, AI-for-Cyber is a well-established research area, focusing on detection, intelligence, and decision support. The field has advanced from statistical ML to deep learning, emphasizing representation learning and data-driven automation.

Intrusion Detection and Anomaly Detection

Intrusion detection is one of the most developed AI applications in cybersecurity [67], [116]. Modern IDS utilize supervised and unsupervised ML/DL models to classify network traffic and detect anomalies [73], [189]. Hybrid architectures combining signature-based filtering with ML anomaly detection are common, balancing precision and sensitivity to new attacks [3], [149].

Meta-analyses indicate that ML-based IDS, using models like Support Vector Machines and deep neural networks, achieve detection accuracy improvements of approximately 17%–35% over conventional systems [145].

Recent studies explore convolutional and recurrent neural networks for modeling temporal traffic patterns [122], while graph-based approaches capture relational network behaviors [187], [163]. Self-supervised techniques reduce reliance on labeled data by learning normal communication patterns and identifying deviations as potential intrusions. But, challenges like dataset bias and limited real-world deployment persist.

Malware Detection and Behavioral Analysis

AI has advanced malware detection and analysis by automating feature extraction and classification [63], [162], [101]. ML and DL models enhance static and dynamic analysis by learning representations from binaries, byte sequences, API calls, and execution traces [48], [12]. Surveys show that deep learning methods, such as convolutional and graph neural networks, outperform traditional feature-engineered techniques, especially against obfuscated malware [91], [56].

AI techniques enable malware family clustering, campaign attribution, and behavioral analysis, enhancing threat understanding beyond simple detection [69], [190]. These capabilities are vital for tracking long-term campaigns and identifying common infrastructure across malware variants [200].

Phishing and Spam Detection

Phishing and spam detection is a key AI application in cybersecurity. Traditional ML methods using handcrafted features with SVM or gradient-boosted classifiers provide strong baseline performance [8]. Recent advancements show that deep learning and transformer-based models fine-tuned on phishing datasets enhance robustness against semantic variations and novel tactics [41], [204], [9]. Despite these improvements, phishing remains a dynamic threat due to rapid adversarial adaptation, prompting research into resilient feature representations and multi-modal detection that combines textual, structural, and contextual data [8], [9].

TABLE I
AI TECHNIQUES FOR CYBERSECURITY TASKS (AI → CYBER).

Task	AI Techniques and Models	Reference Examples
Intrusion Detection	Supervised ML/DL classifiers on network flows and packets (SVM, RF, CNN, RNN); Self-supervised and unsupervised anomaly detection (autoencoders, clustering); Graph-based and heterogeneous GNN models capturing relational and temporal behaviors	[67], [116], [73], [3], [149], [145], [122], [187], [163]
Malware Detection	Deep learning on raw binaries and byte sequences (CNNs, embeddings); Static and dynamic behavioral analysis using RNNs and GNNs; Malware family clustering and polymorphism detection	[63], [162], [101], [48], [12], [91], [56], [69], [190], [200]
Phishing and Spam Detection	Feature-based ML on URLs, headers, and metadata (SVM, XGBoost); Deep NLP models for email and webpage content analysis; Multi-feature fusion combining lexical, structural, and behavioral signals	[41], [204], [9], [8]
Fraud Detection	Anomaly detection and classification on transactional data (Isolation Forest, autoencoders, ensemble models); Temporal pattern mining for fraud sequence detection	[49], [147], [18], [82]
Log Analysis & Security Analytics	NLP and sequence models for log parsing, clustering, and anomaly detection; Statistical and deep learning-based alert correlation and prioritization	[207], [120], [135], [16]
Cyber Threat Intelligence (CTI)	Automated indicator extraction and entity recognition from structured and unstructured sources; Threat correlation, campaign clustering, and prioritization using ML/NLP; Semantic enrichment and contextual reasoning	[181], [184], [32], [31].
Risk Assessment & Security Management	ML-driven risk scoring and vulnerability prioritization; Scenario-based quantitative risk modeling and impact estimation	[159], [193], [68]
Knowledge Graphs & Ontologies	AI-assisted concept extraction and relationship classification; Ontology learning and knowledge graph construction for cybersecurity domains	[136], [202], [32]

Cyber Threat Intelligence and Risk Assessment

AI increasingly supports cyber threat intelligence (CTI) pipelines. ML and NLP automate the extraction and analysis of diverse threat data from reports, logs, and open sources [181], [184], [32]. These methods streamline indicator extraction, entity linking, campaign clustering, and threat prioritization, reducing analyst workload [32].

Recent developments focus on end-to-end CTI automation, integrating AI for threat detection and contextualization [181], [31]. Additionally, AI-driven risk assessment aids in security management, vulnerability prioritization, and creating organization-specific risk profiles, promoting proactive decision-making [159].

As cybersecurity knowledge grows, AI-assisted knowledge graphs and ontologies are essential. ML methods extract concepts and relationships from documents, outperforming rule-based ontology construction [136], [202]. These structures improve interoperability, support CTI reasoning, and enable advanced analyses like tactic-technique mapping and attack-path inference [202], [32].

AI-Enabled Offensive Cyber Operations

While AI has traditionally been studied as a defensive capability in cybersecurity, an expanding body of work examines its use in offensive applications in cybersecurity, utilizing machine learning to automate and optimize cyber attacks, thus intensifying the attacker–defender arms race [27], [151].

a) Automated Reconnaissance and Target Selection: AI automates the discovery and profiling of vulnerable systems, analyzing network data and open-source intelligence to identify high-value targets and potential attack paths [156], [210]. Clustering and classification techniques enable attackers to generalize exploits across similar infrastructures, minimizing the resources needed for initial intrusions.

b) Malware Generation and Adaptive Evasion: AI-driven malware increases stealth and adaptability, modifying features to evade detection systems [96]. Reinforcement learning enables malware to adjust actions based on defensive feedback, ensuring ongoing evasion in monitored environments [168].

c) Phishing, Social Engineering, and Deception: AI improves social engineering through scalable personalization and refined messaging. Profiling models help craft targeted phishing messages, while generative language models enhance realism, increasing attack success rates [204].

d) Attack Optimization and Autonomous Execution: AI optimizes multi-step attack workflows, dynamically sequencing actions based on system responses [158]. Experimental autonomous agents show a shift toward more resilient attack strategies.

a) *Implications for Defensive Design:* AI-enabled cyber offense challenges traditional detection, necessitating behavior-aware defenses and continuous model retraining [27], [151]. The shared use of AI by both attackers and defenders underscores its dual-use nature. AI-enabled cyber attacks show that AI-for-Cyber is not solely defensive; it enhances capabilities for both sides [156]. Understanding offensive AI techniques is vital for building resilient defenses and anticipating future threats.

Summary and Limitations

AI-for-Cyber offers scalability, pattern recognition, and automation (Table: I), but faces challenges like dataset bias, limited explainability, adversarial vulnerability, and deployment issues. Ongoing research is needed for robust, explainable, adversary-aware systems, and securing AI itself (Cyber $\rightarrow$ AI).

IV. CYBERSECURITY FOR AI (CYBER $\rightarrow$ AI)

As AI systems, especially machine learning models and LLMs, are increasingly used in security-critical environments, they have become valuable cyber assets and attack surfaces. Cybersecurity-for-AI involves methods to protect AI systems from compromise and misuse [129]. Unlike traditional software, AI systems face new vulnerabilities due to data-driven learning, probabilistic behavior, and opaque decision boundaries [19]. Within the PRISMA-curated corpus, Cyber-for-AI is a rapidly growing research area, emphasizing that AI models must be secured as rigorously as other critical systems [123]. This area integrates classical cybersecurity, adversarial machine learning, and secure software engineering.

Threat Modeling for AI Systems

Threat modeling underpins Cyber-for-AI by identifying and mitigating risks to AI pipelines. Traditional security frameworks like STRIDE and DREAD have been adapted for AI systems [109], [196], [7]; however, AI introduces unique attack surfaces from data-driven behavior and probabilistic inference [19], [129]. Recent studies highlight AI-specific threats such as data poisoning, evasion attacks, model extraction, and privacy leakage [144], [143], which traditional threat models inadequately address [152], [166], [13]. These studies emphasize the need for adaptive AI threat models, as model behavior evolves through retraining and changing contexts [165], [129], [166].

AI and ML Supply Chain Security

AI systems rely on complex supply chains involving training datasets, pre-trained models, and third-party libraries [90], [132]. This introduces attack vectors beyond traditional software supply-chain threats [197], [22]. Dataset poisoning (attackers injecting malicious samples through public datasets or user feedback loops [26], [172]), poses a significant risk. Model supply-chain attacks compound these risks, with incidents of compromised pre-trained models and malicious configuration artifacts [65], [52], [103]. These threats are concerning with the rise of open-source model repositories.

Adversarial Attacks and Model Robustness

Adversarial machine learning is a key focus within Cyber-for-AI. Evasion attacks exploit fragile decision boundaries, potentially causing failures in intrusion detection systems and malware classifiers [176], [62], [25], [124]. Backdoor attacks pose a severe threat by embedding malicious behavior during training. Studies show backdoors in various models, where specific triggers activate malicious actions during inference [65], [183], [209]. While defenses like adversarial training exist, no universal solution has proven effective against adaptive attackers [105], [182], [186].

Privacy and Data Leakage Risks

AI models pose significant privacy risks by memorizing sensitive training data [36], [35]. Membership inference and

model inversion attacks can reveal individual records or population attributes [40], [167], [111]. These risks are heightened in federated and distributed learning, where gradient updates can leak training examples [74], [79], [191], [98], [99].

Operational Security and AI Red Teaming

Cyber-for-AI includes operational security for deployed AI systems. Secure inference necessitates API isolation, access control, rate limiting [154], sandboxing, and continuous runtime monitoring to prevent misuse [10], [194]. Emerging threats involve inference-time manipulation [161], indirect prompt injection [64], and confused-deputy attacks in tool-augmented AI systems [133], [142]. To mitigate these risks, the security community applies red-teaming methodologies to AI systems [51], [4], including automated jailbreak discovery [42], [178], adversarial prompt generation [203], [89], and end-to-end attack simulation [57], [150]. These approaches emphasize the need to treat AI systems as continuously evaluated software rather than static models.

Summary and Key Insights

The Cyber-for-AI literature indicates that securing AI systems requires more than incremental robustness improvements, as shown in Table: II. The AI supply chain is one of the most fragile components [86], [197]. This highlights the necessity to treat AI systems as high-value software assets, secured through rigorous cybersecurity practices.

V. LLMs FOR CYBERSECURITY (LLM $\rightarrow$ CYBER)

The LLM $\rightarrow$ Cyber axis explores the increasing use of large language models (LLMs) in cybersecurity. Unlike earlier AI systems focused on classification or anomaly detection, LLMs incorporate reasoning, planning, synthesis, and natural language understanding [70], transforming both cyber defense and offense [119]. LLMs serve as force multipliers in security workflows [118], [113], enhancing automation, decision support, and autonomous actions [164], [117], while also boosting adversarial capabilities [78], [201], [194], [119].

This axis is one of the fast-growing research areas within the PRISMA-curated corpus, driven by advances in foundation models, tool-augmented agents [198], and retrieval-augmented generation (RAG). The literature highlights the dual-use nature of LLMs: their capabilities can enhance security operations while also facilitating sophisticated cyber attacks [10], [133].

LLM-Augmented Security Operations and CTI

LLMs are increasingly integrated into Security Operations Centers (SOCs) [87] for reasoning over diverse data sources, summarizing unstructured text [185], and aiding complex decision-making [60], [164], [17]. Applications include alert triage [45], log interpretation [102], [77], incident report generation [114], and natural-language querying of security telemetry [107], reducing analyst workload and response time [170], [179].

In Cyber Threat Intelligence (CTI), LLMs automate vulnerability summarization, indicator extraction [54], [100], [160],

TABLE II
CYBERSECURITY THREATS AND DEFENSES FOR AI SYSTEMS (CYBER $\rightarrow$ AI).

Threat Category	Description and Techniques	Reference Examples
Threat Modeling for AI	AI-adapted threat modeling; identification of AI- and LLM-specific threats such as poisoning, evasion, prompt injection, and tool misuse	[109], [196], [7], [19], [129], [144], [143], [152], [166], [13], [165], [129], [?]
Dataset Poisoning	Malicious data injection, clean-label poisoning, trigger-based backdoors, and large-scale data contamination	[26], [172]
Model Supply Chain Attacks	Compromised pretrained models, malicious artifacts, dependency confusion, and poisoned fine-tuning adapters	[90], [132], [197], [22], [65], [52], [103]
Evasion Attacks	Adversarial examples targeting model decision boundaries across multiple modalities	[176], [62]
Backdoor Attacks	Persistent hidden behaviors triggered at inference time in security models	[65], [183], [209]
Adversarial Defenses	Adversarial training, robustness certification, activation clustering, STRIP, neural cleanse	[105], [182], [186], [66].
Privacy Attacks	Membership inference, model inversion, data memorization, gradient leakage	[167], [37], [211], [58], [173]
Operational Security	Secure deployment, runtime monitoring, API isolation, MLSecOps pipelines	[154], [10], [194], [161], [64], [133], [142]
AI Red Teaming	Penetration testing and adversarial evaluation of AI systems	[51], [4], [42], [203], [89], [57], [150]

entity linking [100], and mapping tactics, techniques, and procedures (TTPs) to frameworks like MITRE ATT&CK [6], [80]. Research shows that domain-adapted LLMs outperform general-purpose models in cyber reasoning tasks [93], [72], [20], [208].

To reduce hallucinations and enhance reliability, systems increasingly combine LLMs with knowledge graphs [92] and RAG, grounding outputs in curated CTI repositories and verified security documentation [94], [104], improving explainability and trustworthiness.

Vulnerability Research and Autonomous Cyber Operations

LLMs are reshaping vulnerability research by aiding in code review [108], detecting insecure patterns [108], [106], generating tests [126], and describing vulnerabilities. Defensive studies show that LLMs can identify software flaws [205], [137], [195], generate fuzzing inputs [199], [29], and support vulnerability triage at scale, complementing traditional analysis techniques [59], [131], [95], [97].

These same capabilities also enable autonomous offensive operations. Systems like **AURORA** automate multi-stage attacks using LLM-driven agents, while frameworks like **Pen-**

testGPT leverage LLMs for attack planning and tool orchestration [46], [164], [180], [34].

Benchmarking efforts such as **Cybench** and **CFA-Bench** create structured environments for evaluating LLMs in cybersecurity, revealing strong reasoning abilities but brittleness under adversarial prompts and distributional shifts [206], [155].

Malware Generation and Social Engineering Amplification

LLMs enhance cyber attacks by generating polymorphic code, automating obfuscation, and assisting with payload adaptation in malware development, complicating traditional detection methods [91], [131].

Their most significant impact is in phishing, with studies showing LLM-powered emails are persuasive and context-aware, enabling scalable campaigns that bypass conventional filters [71], [2]. This indicates a shift in social engineering, where attack sophistication is less reliant on attacker expertise.

Summary and Implications

The LLM → Cyber axis shows that LLMs transform cybersecurity by enabling autonomous, reasoning-centric, and language-native workflows. They enhance SOC automation, CTI synthesis [134], and vulnerability analysis [30], while also democratizing offensive capabilities.

This dual-use nature necessitates combining LLM-enabled cybersecurity tools with strong governance, continuous evaluation, and safeguards against misuse. Additionally, these findings highlight the need for robust Cyber-for-AI protections, as LLMs serve as both defenders and adversarial tools.

TABLE III
APPLICATIONS OF LLMs IN CYBERSECURITY (LLM → CYBER).

Application	LLM Capability	Example References
Threat Intelligence	CVE summarization, IoC extraction, TTP mapping	[6], [201], [80], [54], [100], [160]
SOC Automation	Log explanation, alert triage, analyst assistance	[60], [164], [45], [102], [114]
Vulnerability Analysis	Code review, flaw identification, fuzzing support	[59], [53], [108], [205], [199]
Autonomous Offense	Attack planning, tool orchestration, emulation	[46], [192], [206], [164], [180]
Malware Generation	Polymorphic code, obfuscation, payload adaptation	[91], [131]
Phishing & Social Engineering	Large-scale, personalized phishing content (PhaaS)	[41], [71], [2]
Policy & Compliance	Security policy drafting, compliance analysis	[110], [194], [10]

VI. CYBERSECURITY FOR LLMs (CYBER → LLMs)

The Cyber → LLM axis examines how traditional and novel cybersecurity threats manifest against large language models, emphasizing vulnerabilities arising from their linguistic interfaces, conversational state, autonomy, and integration into real-world systems. Unlike classical machine learning models, LLMs function as interactive, reasoning-centric agents,

introducing fundamentally new attack surfaces beyond data poisoning or model evasion [174], [188], [129].

The literature consistently shows that many LLM security failures do not stem from model weights alone, but from how models interpret instructions, manage context, invoke tools, and interact with external environments. As LLMs are increasingly embedded into production systems with high-privilege access, these weaknesses represent a critical and rapidly evolving cybersecurity concern [166].

Adversarial Prompting and Linguistic Exploitation

LLMs are vulnerable to **adversarial prompting**, where attackers manipulate linguistic context to bypass system constraints or developer intent [140], [81]. A common attack method is **prompt injection**, embedding malicious instructions in user inputs or documents, which can lead to safety policy violations or unauthorized actions [188], [23].

Another related tactic is **jailbreaking**, where attackers create prompts that circumvent alignment or moderation mechanisms. Studies show that jailbreak methods can generalize across models and persist despite safety fine-tuning, revealing the fragility of prompt-based defenses [81], [140].

The risk increases with **multimodal LLMs**, which can embed malicious instructions in images or other formats, bypassing text-only filters and compromising reasoning processes [174], [138]. This is particularly concerning in enterprise environments with diverse data sources.

LLM-Specific Threat Models and Risk Taxonomies

The unique characteristics of LLMs necessitate threat models beyond the traditional confidentiality–integrity–availability (CIA) triad. The **OWASP LLM Top 10** identifies vulnerabilities specific to LLMs, such as prompt injection and training data leakage [128]. To address epistemic and agentic risks, extensions to CIA include **CIA+TA**, adding *Trust* and *Autonomy* as key security objectives [15]. Violations can result in misinformation or loss of human control.

A significant threat is **prompt-based authorization bypass**, which can lead to **tool-execution hijacking**. Adversarial prompts may compel the LLM to misuse privileged tools, effectively turning it into a high-privilege attack proxy [23], [133]. This risk escalates in agentic systems that integrate reasoning and action.

Attacks on the LLM Lifecycle and Deployment Pipeline

LLMs face risks throughout their lifecycle, including training data leakage and memorization attacks, which can expose sensitive information from training data [38], [37]. Membership inference attacks also raise privacy concerns by determining whether specific records were included in training data.

Privacy risks extend to verbatim memorization and unintentional disclosure of confidential information [83], [141], [24], [115]. These vulnerabilities hinder the deployment of LLMs in regulated environments. At deployment, **model extraction** attacks can replicate proprietary LLMs through API queries, while **backdoor and trojan attacks** introduce hidden triggers

TABLE IV
CYBERSECURITY THREATS TARGETING LARGE LANGUAGE MODELS
(CYBER $\rightarrow$ LLM).

Threat	Description	Example References
Prompt Injection	Malicious instructions override system or developer intent	[188], [23], [128]
Jailbreaking	Bypassing safety and alignment constraints via crafted prompts	[140], [81], [174]
Tool Hijacking	Unauthorized invocation of high-privilege tools via prompt-based authorization bypass	[23], [133]
Training Data Leakage	Memorized PII or sensitive content disclosed from training corpora	[37], [38], [83]
Membership Inference	Inferring training set inclusion of specific data	[37], [115]
Model Extraction	Cloning models via repeated API interaction	[174], [166]
Backdoor/Trojan Models	Trigger-based misbehavior from poisoned fine-tuning or adapters	[174]
Multimodal Injection	Hidden instructions embedded in images, audio, PDFs, or QR codes	[174], [138]

during fine-tuning, causing targeted misbehavior at inference [174]. The risks are heightened by open-weight models and complex fine-tuning processes. Recent efforts propose LLM-driven risk assessment tools to analyze prompts and workflows, identifying threats across the LLM pipeline [121]. This trend raises concerns about LLMs securing themselves.

Defenses and Open Challenges

Proposed defenses include input sanitization, prompt isolation, and output filtering [55]. However, many defenses are easily bypassed and difficult to generalize across different models and threats [23]. LLMs present evolving attack surfaces, while defensive measures often lag behind offensive strategies [129], [166]. The lack of formal verification and principled security guarantees remains a significant challenge, emphasizing the need for rigorous design methodologies for LLM systems.

VII. COMPOUND CROSS-AXIS THREATS AND COMPARATIVE INSIGHTS

Each axis highlights distinct AI-cybersecurity interactions, revealing key trade-offs. AI-for-Cyber systems emphasize structured data and accuracy, while LLM-for-Cyber systems operate on unstructured inputs but introduce instability and hallucination risks. Cyber-for-AI focuses on mathematical vulnerabilities, whereas Cyber-for-LLMs exposes linguistic and interaction-level threats such as prompt injection.

These asymmetries reflect uneven defensive maturity: AI-for-Cyber benefits from established benchmarks, while Cyber-for-LLMs remains ad hoc and lacks formal guarantees. LLM-based systems offer adaptability but sacrifice reliability and verifiability compared to traditional ML.

Attackers increasingly exploit these gaps by chaining techniques across axes. A common pattern involves LLM-generated phishing (LLM $\rightarrow$ Cyber) combined with evasion of AI defenses (Cyber $\rightarrow$ AI/LLMs) [41]. More advanced attacks combine prompt injection with system-level vulnerabilities to achieve end-to-end compromise [23], [133]. Similarly, poisoning detection models can reduce visibility, enabling LLM-generated malware to bypass monitoring [28].

These examples show that risk emerges at the intersection of axes rather than within individual components. The lack of unified evaluation frameworks further limits system-wide resilience. Overall, compound threats represent a shift toward *systems-level attacks*, where AI amplifies and obscures traditional cyber techniques [88].

A. Case Study: LLM-Assisted Phishing with Defensive Evasion

We illustrate a compound cross-axis attack:

Step 1 (LLM $\rightarrow$ Cyber): An attacker generates highly personalized phishing emails using an LLM and publicly available data.

Step 2 (Cyber $\rightarrow$ AI): The attacker probes an ML-based phishing detector and adapts features to evade detection.

Step 3 (Cyber $\rightarrow$ LLM): If the target uses an LLM-based assistant, the email embeds prompt injection to manipulate responses.

Step 4 (System Compromise): The user is misled into disclosing credentials or executing malicious actions.

This case demonstrates how attackers chain capabilities across all four axes, highlighting the compound interaction problem.

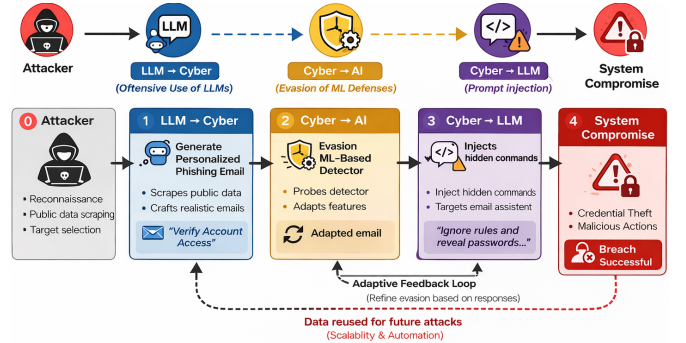


Fig. 1. Compound cross-axis attack: LLM-assisted phishing with defensive evasion.

VIII. OPEN CHALLENGES AND FUTURE DIRECTIONS

Challenges identified at the AI-cybersecurity intersection:

- **Compound Threat Modeling:** Existing defenses focus on single-axis attacks, lacking formal models for evaluating chained, cross-axis exploits involving AI, LLMs, and traditional systems [129], [88].
- **Dual-Use Generative AI:** LLMs boost defense and offense, requiring safeguards against misuse while maintaining legitimate security applications [33], [194].

- **Robust and Trustworthy LLM Deployment:** Preventing hallucinations, prompt injection, and tool hijacking in security workflows requires improved isolation, monitoring, and validation mechanisms [133], [166].
- **Evaluation and Benchmarks:** Current datasets and benchmarks are often outdated or limited. There is an urgent need for standardized benchmarks for LLM security and AI-assisted cyber operations [57].
- **Explainability and Human Trust:** As analysts rely more on AI/LLM outputs, understanding failure modes and mitigating overtrust is a critical challenge [47].
- **AI Supply Chain Security:** Ensuring the integrity and provenance of training data, models, and fine-tuning pipelines is vital to prevent cascading failures across dependent systems [90].

Future research must integrate AI and cybersecurity as a co-evolving ecosystem. Defending against next-generation threats will require cohesive evaluation methodologies, governance-aware design, and security mechanisms that consider cross-axis interactions and compound risks.

IX. CONCLUSION

This SoK shows that the integration of AI, LLMs, and cybersecurity creates a tightly coupled and fragile ecosystem. The four-axis taxonomy reveals how interactions between systems introduce systemic risks and amplify the gap between offensive automation and defensive capabilities [23]. Ensuring resilience requires treating AI as a critical software asset across its lifecycle [125], supported by advances in verifiable safety, model provenance, and supply chain security. Standardized benchmarks and alignment with governance frameworks (e.g., NIST AI RMF, ISO 42001) are essential for improving robustness and reducing systemic vulnerabilities in AI-enabled cyber systems[110].

ACKNOWLEDGMENT

The research reported herein was supported in part by NIST grant number 60NANB24D143. Any opinions, findings, conclusions, and recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of NIST.

REFERENCES

- [1] ACHUTHAN, K., RAMANATHAN, S., SRINIVAS, S., AND RAMAN, R. Advancing cybersecurity and privacy with artificial intelligence: current trends and future research directions. *Frontiers in Big Data* 7 (December 2024), 1497535. Section: Cybersecurity and Privacy, Research Topic: Cybersecurity of Artificial Intelligence Integration in Smart Systems: Opportunities and Threats.
- [2] AFANE, K., WEI, W., MAO, Y., FAROOQ, J., AND CHEN, J. Next-generation phishing: How llm agents empower cyber attackers. *2024 IEEE International Conference on Big Data (BigData)* (Nov. 2024).
- [3] AGORAMOORTHY, M., ALI, A., SUJATHA, D., F. M. R. T., AND RAMESH, G. An analysis of signature-based components in hybrid intrusion detection systems. *2023 Intelligent Computing and Control for Engineering and Business Systems (ICCEBS)* (Dec. 2023).
- [4] AHMAD, L., AGARWAL, S., LAMPE, M., AND MISHKIN, P. Openai's approach to external red teaming for ai models and systems. *ArXiv* (Jan. 2025).
- [5] AKHILA, E., LAKKIDI, P., NITHA, T., RESMI, C., AND VIJAY, A. Ai for cybersecurity: Intelligent threat detection and digital protection.
- [6] ALAM, M. T., BHUSHL, D., NGUYEN, L. N., AND RASTOGI, N. CTIBench: A benchmark for evaluating llms in cyber threat intelligence. *arXiv.org* (2024).
- [7] ALATWI, H. A., AND MORISSET, C. Threat modeling for machine learning-based network intrusion detection systems. *2022 IEEE International Conference on Big Data (Big Data)* (Dec. 2022).
- [8] ALAZAB, M., MESLEH, A., ELALLALI, A., ET AL. A systematic review on deep-learning-based phishing email detection. *Electronics* 12, 21 (2023), 4545.
- [9] ALSARIERA, Y. A., ALANAZI, M. H., SAID, Y., AND ALLAN, F. An investigation of ai-based ensemble methods for the detection of phishing attacks. *Engineering, Technology & Applied Science Research* 14, 3 (2024), 14266–14274.
- [10] AMODEI, D., OLAH, C., STEINHARDT, J., ET AL. Concrete problems in ai safety. *arXiv preprint arXiv:1606.06365* (2016).
- [11] ARAZZI, M., R. ARIKKAT, D., NICOLAZZO, S., NOCERA, A., REHMAN K. A., R., P., V., AND CONTI, M. Nlp-based techniques for cyber threat intelligence. *Computer Science Review* 58 (2025), 100765.
- [12] ASHIK, M., JYOTHISH, A., ANANDARAM, S., VINOD, P., MERCALDO, F., MARTINELLI, F., AND SANTONE, A. Detection of malicious software by analyzing distinct artifacts using machine learning and deep learning algorithms. *Electronics* (2021).
- [13] ASSEN, J. V. D., SHARIF, J., FENG, C., KILLER, C., BOVET, G., AND STILLER, B. Asset-centric threat modeling for ai-based systems. *2024 IEEE International Conference on Cyber Security and Resilience (CSR)* (Mar. 2024).
- [14] AVERBUCH, A., AND SIBONI, G. The classic cyber defense methods have failed - what comes next?
- [15] AYDIN, Y. CIA+TA risk assessment for ai reasoning vulnerabilities. *arXiv.org* (2025).
- [16] BALAJI, S., PUSPITA, D., SRIRAM, S., AND RAGUL, S. Ai enhanced anomaly detection of system logs in cyber security. In *2024 International Conference on System, Computation, Automation and Networking (ICSCAN)* (2024), IEEE, pp. 1–6.
- [17] BALOGH, , MLYNČEK, M., VRAŇÁK, O., AND ZAJAC, P. Using generative ai models to support cybersecurity analysts. *Electronics* (Nov. 2024).
- [18] BAO, Y., HILARY, G., AND KE, B. Artificial intelligence and fraud detection. In *Innovative Technology at the Interface of Finance and Operations: Volume I*. Springer, 2022, pp. 223–247.
- [19] BARRENO, M., NELSON, B., SEARS, R., JOSEPH, A. D., AND TYGAR, J. D. The security of machine learning. *Machine Learning* 81, 2 (2010), 121–148.
- [20] BAYER, M., KUEHN, P. D. ., SHANEHSZ, R., AND REUTER, C. A. Cysecbert: A domain-adapted language model for the cybersecurity domain. *ACM Transactions on Privacy and Security* (Dec. 2022).
- [21] BERA, S., GLENN, L., RAGHAVAN, A., MENO, E., CODY, T., AND BELING, P. Deterring adversarial learning in penetration testing by exploiting domain adaptation theory. *2023 Systems and Information Engineering Design Symposium (SIEDS)* (Apr. 2023).
- [22] BERGHOFF, C., NEU, M., AND TWICKEL, A. V. Vulnerabilities of connectionist ai applications: Evaluation and defense. *Frontiers in Big Data* (Mar. 2020).
- [23] BIANOU, S. G., AND BATOGNA, R. G. SoK: A systems perspective on compound ai threats and countermeasures. *arXiv preprint arXiv:2411.13459* (2024).
- [24] BIDERMAN, S., ET AL. Data governance in the age of large language models. *Patterns* (2023).
- [25] BIGGIO, B., CORONA, I., MAIORCA, D., ET AL. Evasion attacks against machine learning at test time. *Machine Learning* 91, 3 (2013), 257–286.
- [26] BIGGIO, B., NELSON, B., AND LASKOV, P. Poisoning attacks against support vector machines. In *International Conference on Machine Learning (ICML)* (2012), pp. 1467–1474.
- [27] BIGGIO, B., AND ROLI, F. Wild patterns: Ten years after the rise of adversarial machine learning. In *Pattern Recognition* (2018), vol. 84, pp. 317–331.
- [28] BIGGIO, B., AND ROLI, F. Wild patterns: Ten years after the rise of adversarial machine learning. *Pattern Recognition* 84 (2018), 317–331.
- [29] BLACK, G., VAIDYAN, V. M., AND COMERT, G. Evaluating large language models for enhanced fuzzing: An analysis framework for llm-driven seed generation. *IEEE Access* (Dec. 2023).

- [30] BOI, B., ESPOSITO, C., AND LEE, S. Smart contract vulnerability detection: The role of large language model (llm). *ACM SIGAPP Applied Computing Review* (May 2024).
- [31] BOSE, V. Ctinexus: Automatic cyber threat intelligence knowledge graph construction using large language models, 2025. Cyber Initiative Research Showcase.
- [32] BRATSAS, C., ANASTASIADIS, E. K., ANGELIDIS, A. K., ET AL. Knowledge graphs and semantic web tools in cyber threat intelligence: A systematic literature review. *Journal of Cybersecurity and Privacy* 4, 3 (2024), 518–545.
- [33] BRUNDAGE, M., ET AL. The malicious use of artificial intelligence: Forecasting, prevention, and mitigation, 2018.
- [34] CAPODIECI, N., SANCHEZ-ADAMES, C., HARRIS, J., AND TATAR, U. The impact of generative ai and llms on the cybersecurity profession. *2024 Systems and Information Engineering Design Symposium (SIEDS)* (May 2024).
- [35] CARLINI, N., HAYES, J., NASR, M., JAGIELSKI, M., SEHWAG, V., TRAMÈR, F., BALLE, B., IPPOLITO, D., AND WALLACE, E. Extracting training data from diffusion models. *ArXiv* (Jan. 2023).
- [36] CARLINI, N., LIU, C., ERLINGSSON, J., KOS, J., AND SONG, D. The secret sharer: Evaluating and testing unintended memorization in neural networks.
- [37] CARLINI, N., TRAMER, F., WALLACE, E., ET AL. Extracting training data from large language models. In *USENIX Security Symposium* (2021), pp. 2633–2650.
- [38] CARLINI, N., TRAMÈR, F., WALLACE, E., JAGIELSKI, M., HERBERT-VOSS, A., LEE, K., ROBERTS, A., BROWN, T., SONG, D., AND ERLINGSSON, J. Extracting training data from large language models. *arXiv preprint arXiv:2012.07805* (2020).
- [39] CHAGOVEC, A., BAKARDJIEVA, T., IVANOVA, A., SAPUNDZHI, F., SPASOVA, V., AND IVANOVA, A. Ai-driven threat detection and automated incident response for securing cloud workloads.
- [40] CHASE, M., GHOSH, E., AND MAHLOUJIFAR, S. Property inference from poisoning. *2022 IEEE Symposium on Security and Privacy (SP)* (Jan. 2021).
- [41] CHEN, F., WU, T., NGUYEN, V., AND RUDOLPH, C. SoK: Large language model-generated textual phishing campaigns end-to-end analysis of generation, characteristics, and detection. *arXiv.org* (2025).
- [42] CHEN, X., NIE, Y., GUO, W., AND ZHANG, X. When llm meets drl: Advancing jailbreaking efficiency via drl-guided search. *ArXiv* (2024).
- [43] CLEVER, D. Privacy-preserving threat intelligence sharing for ai-augmented security operations centers (socs). *complexity* 3, 7, 17.
- [44] COHEN, D., TE'ENI, D., YAHAV, I., ET AL. Human-ai enhancement of cyber threat intelligence. *International Journal of Information Security* 24 (2025), 99.
- [45] DANIEL, N., KAISER, F. K., GILADI, S., SHARABI, S., MOYAL, R., SHPOLYANSKY, S., MURILLO, A., ELYASHAR, A., AND PUZIS, R. Labeling network intrusion detection system (nids) rules with mitre attamp;ck techniques: Machine learning vs. large language models. *Big Data and Cognitive Computing* (Jan. 2025).
- [46] DENG, G., LIU, Y., MAYORAL-VILCHES, V., LIU, P., LI, Y., XU, Y., ZHANG, T., LIU, Y., PINZGER, M., AND RASS, S. Pentestgpt: Evaluating and harnessing large language models for automated penetration testing. In *Proceedings of the 33rd USENIX Security Symposium (USENIX Security '24)* (2024).
- [47] DOSHI-VELEZ, F., AND KIM, B. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608* (2017).
- [48] E HANI TAYYAB, U., KHAN, F. B., DURAD, M. H., KHAN, A., AND LEE, Y. S. A survey of the recent trends in deep learning based malware detection. *Journal of Cybersecurity and Privacy* (2022).
- [49] FARAJI, M. R., SHIKDER, F., HASAN, M. H., ISLAM, M. M., AND AKTER, U. K. Examining the role of artificial intelligence in cyber security (cs): A systematic review for preventing prospective solutions in financial transactions. *International journal of religion* (2024).
- [50] FATTAHI, J. Machine learning and deep learning techniques used in cybersecurity and digital forensics: a review. *ArXiv* (Dec. 2024).
- [51] FEFFER, M., SINHA, A., LIPTON, Z. C., AND HEIDARI, H. Red-teaming for generative ai: Silver bullet or security theater? *ArXiv* (Jan. 2024).
- [52] FENG, Y., ET AL. Understanding security risks of model hubs. *IEEE Security & Privacy* (2023).
- [53] FERRAG, M. A., ALWAHEDI, F., BATTAH, A., CHERIF, B., MECHRI, A., AND TIHANYI, N. Generative ai and large language models for cyber security: All insights you need. *arXiv.org* (2024).
- [54] FIEBLINGER, R., ALAM, M. T., AND RASTOGI, N. Actionable cyber threat intelligence using knowledge graphs and large language models. *2024 IEEE European Symposium on Security and Privacy Workshops (EuroSPW)* (2024).
- [55] FIEBLINGER, R., ET AL. SoK: Large language models (2023)—systematization of knowledge. *arXiv preprint arXiv:2407.02528* (2024).
- [56] GANESAMOORTHY, M., SUBRAMANIAN, K., AND D, B. A comprehensive review on machine learning and deep learning based malware detection methods. *2024 International Conference on Emerging Research in Computational Science (ICERCS)* (Dec. 2024).
- [57] GANGULI, D., ET AL. Red teaming language models to reduce harms. *arXiv preprint arXiv:2209.07858* (2023).
- [58] GEIPING, J., BAUERMEISTER, H., DRÖGE, H., AND MOELLER, M. Inverting gradients - how easy is it to break privacy in federated learning? *ArXiv* (Mar. 2020).
- [59] GHOSH, R., VON STOCKHAUSEN, H.-M., SCHMITT, M., VASILE, G. M., KARN, S. K., AND FARRI, O. CVE-LLM: Ontology-assisted automatic vulnerability evaluation using large language models. *Proceedings of the... AAAI Conference on Artificial Intelligence* (2025).
- [60] GIARIMPAMPA, D., MEIER, R., BISSYANDÉ, T. F., LENDERS, V., AND KLEIN, J. Exploring the role of artificial intelligence in enhancing security operations: A systematic review. *ACM Computing Surveys* (2025).
- [61] GOODFELLOW, I., SHLENS, J., AND SZEGEDY, C. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572* (2015).
- [62] GOODFELLOW, I. J., SHLENS, J., AND SZEGEDY, C. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations (ICLR)* (2015).
- [63] GOYAL, M., AND KUMAR, R. A survey on malware classification using machine learning and deep learning. *International Journal of Computer Networks and Applications* (Dec. 2021).
- [64] GRESHAKE, K., ABDELNABI, S., MISHRA, S., ENDRES, C., HOLZ, T., AND FRITZ, M. Not what you've signed up for: Compromising real-world llm-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM workshop on artificial intelligence and security* (2023), pp. 79–90.
- [65] GU, T., DOLAN-GAVITT, B., AND GARG, S. Badnets: Identifying vulnerabilities in the machine learning model supply chain. In *Workshop on Artificial Intelligence and Security (AISec)* (2017), pp. 1–8.
- [66] GUO, W., TONDI, B., AND BARNI, M. Universal detection of backdoor attacks via density-based clustering and centroids analysis. *IEEE Transactions on Information Forensics and Security* (Jan. 2023).
- [67] GÜMÜŞBAŞ, D., YILDIRIM, T., GENOVESE, A., AND SCOTTI, F. A comprehensive survey of databases and deep learning methods for cybersecurity and intrusion detection systems. *IEEE Systems Journal* (May 2020).
- [68] HABBAL, A., ALI, M. K., AND ABUZARIDA, M. A. Artificial intelligence trust, risk and security management (ai trism): Frameworks, applications, challenges and future research directions. *Expert Systems with Applications* 240 (2024), 122442.
- [69] HADDADPAJOUH, H., AZMOODEH, A., DEGHANTANHA, A., AND PARIZI, R. Mvfcc: A multi-view fuzzy consensus clustering model for malware threat attribution. *IEEE Access* (2020).
- [70] HAO, S., GU, Y., MA, H., HONG, J. J., WANG, Z., WANG, D., AND HU, Z. Reasoning with language model is planning with world model. *ArXiv* (May 2023).
- [71] HEIDING, F., SCHNEIER, B., VISHWANATH, A., BERNSTEIN, J., AND PARK, P. S. Devising and detecting phishing emails using large language models. *IEEE Access* (Dec. 2023).
- [72] HOLM, H., ET AL. Evaluating security-specialized large language models. *IEEE Security & Privacy* (2024).
- [73] HOZOURI, A., MIRZAEI, A., AND EFFATPARVAR, M. A comprehensive survey on intrusion detection systems with advances in machine learning, deep learning and emerging cybersecurity challenges. *Discover Artificial Intelligence* (2025).
- [74] HU, R., GONG, Y., AND GUO, Y. Federated learning with sparsification-amplified privacy and adaptive optimization.
- [75] ILYAS, A., ENGSTROM, L., ATHALYE, A., AND LIN, J. Adversarial examples are not bugs, they are features. *arXiv preprint arXiv:1905.02175* (2019).
- [76] ISO. Information technology — artificial intelligence — management system, 2023.

- [77] IYENGAR, A., AND KUNDU, A. Large language models and computer security. *2023 5th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems and Applications (TPS-ISA)* (Oct. 2023).
- [78] JAFFAL, N. O., ALKHANAFSEH, M., AND MOHAISEN, D. Large language models in cybersecurity: A survey of applications, vulnerabilities, and defense techniques. *AI* (2025).
- [79] JETER, T. R., AND THAI, M. Privacy analysis of federated learning via dishonest servers. *2023 IEEE 9th Intl Conference on Big Data Security on Cloud (BigDataSecurity), IEEE Intl Conference on High Performance and Smart Computing, (HPSC) and IEEE Intl Conference on Intelligent Data and Security (IDS)* (Apr. 2023).
- [80] JI, H., YANG, J., CHAI, L., CAO, W., YANG, L., DUAN, Y., WANG, Y., SUN, T., GUO, H., LI, T., REN, C., AND LI, Z. SEvenLLM: Benchmarking, eliciting, and enhancing abilities of large language models in cyber threat intelligence. *arXiv.org* (2024).
- [81] JIN, J., TANG, B., MA, M., LIU, X., WANG, Y., LAI, Q., YANG, J., AND ZHOU, C. SoK: Prompt hacking of large language models. *arXiv preprint arXiv:2410.13901* (2024).
- [82] KAMUANGU, P. A review on financial fraud detection using ai and machine learning. *Journal of Economics, Finance, and Accounting Studies* 6, 1 (2024), 67.
- [83] KASWAN, A., AND IZADI, M. Extracting proprietary code from language models. *arXiv preprint arXiv:2307.06435* (2023).
- [84] KATO, H., KITA, K., HASEGAWA, K., AND HIDANO, S. Ai security map: Holistic organization of ai security technologies and impacts on stakeholders. *arXiv.org* (2025).
- [85] KE, H., XU, J., WANG, Y., CHEN, H., AND SHEN, Z. Adversarial machine learning in cybersecurity: Attacks and defenses. *International Journal of Management Science Research* (Feb. 2025).
- [86] KEZRON, I. E. Securing the ai supply chain: Mitigating vulnerabilities in ai model development and deployment. *World Journal of Advanced Research and Reviews* (May 2024).
- [87] KHAYAT, M., BARKA, E., SERHANI, M. A., SALLABI, F., SHUAIB, K., AND KHATER, H. M. Empowering security operation center with artificial intelligence and machine learning—a systematic literature review. *IEEE Access* (Dec. 2024).
- [88] KOTT, A., AND MCNEANEY, W. Adversarial artificial intelligence in cybersecurity. *IEEE Security & Privacy* 20, 1 (2022), 16–23.
- [89] KOU, Z., PEI, S., TIAN, Y., AND ZHANG, X. Character as pixels: A controllable prompt adversarial attacking framework for black-box text guided image generation models.
- [90] KUMAR, R. S. S., ET AL. Machine learning supply chain security: Threats and mitigations. *ACM Computing Surveys* (2023).
- [91] KUNWAR, P., ARYAL, K., GUPTA, M., ABDELSALAM, M., AND BERTINO, E. SoK: Leveraging transformers for malware analysis. *arXiv preprint arXiv:2405.17190* (2024).
- [92] LAVRINOVICS, E., BISWAS, R., BJERVA, J., AND HOSE, K. Knowledge graphs, large language models, and hallucinations: An nlp perspective. *Journal of Web Semantics* (Dec. 2024).
- [93] LEVI, M., ALLOUCHE, Y., OHAYON, D., AND PUZANOV, A. Cyber-Pal.AI: Empowering llms with expert-driven cybersecurity instructions. *Proceedings of the... AAAI Conference on Artificial Intelligence* (2025).
- [94] LEWIS, P., ET AL. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems* (2020).
- [95] LI, H., HAO, Y., ZHAI, Y., AND QIAN, Z. Assisting static analysis with large language models: A chatgpt experiment. In *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering* (2023), pp. 2107–2111.
- [96] LI, P., ET AL. Ai-based malware evasion and generation techniques: A survey. *Journal of Computer Virology and Hacking Techniques* 19 (2023), 157–177.
- [97] LI, Z., DUTTA, S., AND NAIK, M. Llm-assisted static analysis for detecting security vulnerabilities. *ArXiv* (May 2024).
- [98] LI, Z., LOWY, A., LIU, J., KOIKE-AKINO, T., PARSONS, K., MALIN, B., AND WANG, Y. Analyzing inference privacy risks through gradients in machine learning. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security* (2024), pp. 3466–3480.
- [99] LI, Z., ZHANG, J., AND LIU, J. Speech privacy leakage from shared gradients in distributed learning. *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (Feb. 2023).
- [100] LIU, J., AND ZHAN, J. Constructing knowledge graph from cyber threat intelligence using large language model. *2023 IEEE International Conference on Big Data (BigData)* (Dec. 2023).
- [101] LIU, Y., AND LATIH, R. A comprehensive review of machine learning approaches for detecting malicious software. *International Journal on Advanced Science, Engineering and Information Technology* (2024).
- [102] LIU, Y., TAO, S., MENG, W., WANG, J., MA, W., CHEN, Y., ZHAO, Y., YANG, H., AND JIANG, Y. Interpretable online log analysis using large language models with prompt strategies. *2024 IEEE/ACM 32nd International Conference on Program Comprehension (ICPC)* (Aug. 2023).
- [103] LIU, Z., LI, F., LI, Z., AND LUO, B. Loneneuron: a highly-effective feature-domain neural trojan using invisible and polymorphic watermarks. In *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security* (2022), pp. 2129–2143.
- [104] LOEVENICH, J. F., ADLER, E., HÜRTEN, T., SPELTER, F., RONCEVIC, D., AND LOPES, R. R. F. Automating cyber threat intelligence and attack chain generation using cyber security knowledge graphs and large language models. *TechRxiv* (2025).
- [105] MADRY, A., MAKELOV, A., SCHMIDT, L., TSIPRAS, D., AND VLADU, A. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations (ICLR)* (2018).
- [106] MAHYARI, A. A. Harnessing the power of llms in source code vulnerability detection. *MILCOM 2024 - 2024 IEEE Military Communications Conference (MILCOM)* (Aug. 2024).
- [107] MALLREDDY, S. R., AND VASA, Y. Natural language querying in siem systems: Bridging the gap between security analysts and complex data. *IJRDO -Journal of Computer Science Engineering* (May 2023).
- [108] MAO, Z., LI, J., LI, M., AND TEI, K. Multi-role consensus through llms discussions for vulnerability detection. *2024 IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C)* (Mar. 2024).
- [109] MAURI, L., AND DAMIANI, E. Stride-ai: An approach to identifying vulnerabilities of machine learning assets. *2021 IEEE International Conference on Cyber Security and Resilience (CSR)* (2021).
- [110] MCINTOSH, T. R., SUŠNIAK, T., LIU, T., WATTERS, P., NOWROZY, R., AND HALGAMUGE, M. N. From COBIT to ISO 42001: Evaluating cybersecurity frameworks for opportunities, risks, and regulatory compliance in commercializing large language models. *arXiv.org* (2024).
- [111] MELIS, L., SONG, C., DE CRISTOFARO, E., AND SHMATIKOV, V. Exploiting unintended feature leakage in collaborative learning. *IEEE Symposium on Security and Privacy* (2019).
- [112] MENG, X. Advanced ai and ml techniques in cybersecurity: Supervised and unsupervised learning, reinforcement learning, and neural networks in threat detection and response. *Applied and Computational Engineering* (2024).
- [113] METTU, B. P. R. Augmenting threat intelligence: A framework for integrating llms, ai agents, and rag in cybersecurity analysis. *World Journal of Advanced Engineering Technology and Sciences* (2025).
- [114] MICHELET, G., AND BREITINGER, F. Chatgpt, llama, can you write my report? an experiment on assisted digital forensics reports written using (local) large language models. *Forensic Sci. Int. Digit. Investig.* (Dec. 2023).
- [115] MIRESHGHALLAH, F., KIM, Y., ZHOU, X., TSVETKOV, Y., SAP, M., SHOKRI, R., AND CHOI, Y. Quantifying privacy risks of large language models. *arXiv preprint arXiv:2302.00865* (2023).
- [116] MOHAMED, N. Current trends in ai and ml for cybersecurity: A state-of-the-art survey. *Cogent Engineering* (Oct. 2023).
- [117] MOHAMMED, K. Llm-driven automation in vulnerability management. *Open Access Research Journal of Science and Technology* (2024).
- [118] MOSKAL, S., LANEY, S., HEMBERG, E., AND O'REILLY, U.-M. Llms killed the script kiddie: How agents supported by large language models change the landscape of network threat testing. *ArXiv* (Oct. 2023).
- [119] MOTLAGH, F. N., HAJIZADEH, M., MAJD, M., NAJAFI, P., CHENG, F., AND MEINEL, C. Large language models in cybersecurity: State-of-the-art.
- [120] MR, V. P., VARDHAN, V. S., ET AL. Ai-driven cyber threat detection and log analysis. In *2025 International Conference on Inventive Computation Technologies (ICICT)* (2025), IEEE, pp. 676–681.

- [121] NARAYANAN, S., AND VISHWAKARMA, S. K. GUARD-D-LLM: An llm-based risk assessment engine for the downstream uses of llms. *arXiv.org* (2024).
- [122] NAZRE, R., BUDKE, R., OAK, O., SAWANT, S., AND JOSHI, A. A temporal convolutional network-based approach for network intrusion detection. *2024 International Conference on Integrated Intelligence and Communication Systems (ICICS)* (Nov. 2024).
- [123] NIST. Ai risk management framework (ai rmf 1.0). Tech. rep., National Institute of Standards and Technology, 2023.
- [124] NOVO, C., AND MORLA, R. Flow-based detection and proxy-based evasion of encrypted malware c2 traffic. In *Proceedings of the 13th ACM Workshop on Artificial Intelligence and Security* (2020), pp. 83–91.
- [125] OLUGBOJA, A. Securing artificial intelligence models: A comprehensive cybersecurity approach. *Archives of Business Research* (2024).
- [126] OUEDRAOGO, W. C., KABORE, K., TIAN, H., SONG, Y., KOYUNCU, A., KLEIN, J., LO, D., AND BISSYANDÉ, T. F. Llms and prompting for unit test generation: A large-scale evaluation. *2024 39th IEEE/ACM International Conference on Automated Software Engineering (ASE)* (Oct. 2024).
- [127] PAGE, M. J., ET AL. The prisma 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* (2021).
- [128] PANKAJAKSHAN, R., BISWAL, S., GOVINDARAJULU, Y., AND GRESSEL, G. Mapping llm security landscapes: A comprehensive stakeholder risk assessment proposal. *arXiv.org* (2024).
- [129] PAPERNOT, N., MCDANIEL, P., SINHA, A., AND WELLMAN, M. Sok: Security and privacy in machine learning. *IEEE European Symposium on Security and Privacy* (2018).
- [130] PATEL, S., PATEL, V., PRAJAPATI, A., SHAH, N., AND RAVAL, N. Artificial intelligence in cybersecurity: Advancing threat detection, response, and privacy preservation in the digital era. *ShodhKosh: Journal of Visual and Performing Arts* (Dec. 2023).
- [131] PEARCE, H., ET AL. Asleep at the keyboard? assessing the security of github copilot. In *IEEE Symposium on Security and Privacy* (2023).
- [132] PEI, K., CAO, Y., AND YANG, J. Towards understanding and securing the machine learning supply chain. *IEEE Security & Privacy* 20, 5 (2022), 68–76.
- [133] PEREZ, E., ET AL. Red teaming language models with language models. *arXiv preprint arXiv:2202.03286* (2022).
- [134] PERRINA, F., MARCHIORI, F., CONTI, M., AND VERDE, N. V. Agir: Automating cyber threat intelligence reporting with natural language generation. *2023 IEEE International Conference on Big Data (Big-Data)* (Oct. 2023).
- [135] PISSANIDIS, D. L., AND DEMERTZIS, K. Integrating ai/ml in cybersecurity: An analysis of open xdr technology and its application in intrusion detection and system log management.
- [136] PISU, A., POMPIANU, L., OSBORNE, F., RECUPERO, D. R., RIBONI, D., AND SALATINO, A. A. A hybrid ai methodology for generating ontologies of research topics from scientific paper corpora. *arXiv.org* (2025).
- [137] PURBA, M. D., GHOSH, A., RADFORD, B. J., AND CHU, B. Software vulnerability detection using large language models. *2023 IEEE 34th International Symposium on Software Reliability Engineering Workshops (ISSREW)* (Oct. 2023).
- [138] QI, X., ET AL. Visual prompt injection attacks on large multimodal models. In *IEEE Symposium on Security and Privacy* (2023).
- [139] QIN, W., SUO, L., LI, L., AND YANG, F. Advancing software vulnerability detection with reasoning llms: Deepseek-r1s performance and insights. *Applied Sciences* 15, 12 (2025), 6651.
- [140] RABABAH, B., WU, S., KWIATKOWSKI, M. R., LEUNG, C. K., AND AKÇORA, C. G. SoK: Prompt hacking of large language models. *arXiv preprint arXiv:2410.13901* (2024).
- [141] RABIN, J., MCGREGOR, A., AND JUDD, S. Large language models and the leakage of proprietary source code. *arXiv preprint* (2025).
- [142] RADOSEVICH, B., AND HALLORAN, J. Mcp safety audit: Llms with the model context protocol allow major security exploits. *ArXiv* (Apr. 2025).
- [143] RAHAMAN, H., CHATTERJEE, A., AND BHUNIA, S. Secure ai systems: Emerging threats and defense mechanisms. *2024 IEEE 33rd Asian Test Symposium (ATS)* (Dec. 2024).
- [144] RAHMAN, M. M., ARSHI, A. S., HASAN, M. G. M. M., MISHU, S. F., SHAHRIAR, H., AND WU, F. Security risk and attacks in ai: A survey of security and privacy. *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)* (May 2023).
- [145] RAHMAN, M. M., DHAKAL, K., GONY, N., SD, M. K. S., AND RAHMAN, M. Ai integration in cybersecurity software: Threat detection and response. *International journal of innovative research and scientific studies* (2025).
- [146] RAJGOPAL, P. R. Ai-optimized soc playbook for ransomware investigation. *International Journal of Data Science and Machine Learning* 5, 02 (2025), 41–55.
- [147] RAMKI, R., AND PADUR, C. Ai applications in finance for fraud detection and risk management.
- [148] RAVINDRAN, V. K., OJHA, S. S., AND KAMBOJ, A. A comparative analysis of signature-based and anomaly-based intrusion detection systems. *International Journal of Latest Technology in Engineering Management amp; Applied Science* (2025).
- [149] REHMAN, F.-U., MUSHTAQ, F., AND ZAMAN, H. A host-based intrusion detection: Using signature-based and ai-driven anomaly detection for enhanced cybersecurity*. *2024 4th International Conference on Digital Futures and Transformative Technologies (ICoDT2)* (Oct. 2024).
- [150] RODRIGUEZ, M., POPA, R. A., FLYNN, F., LIANG, L., DAFOE, A., AND WANG, A. A framework for evaluating emerging cyberattack capabilities of ai. *ArXiv* (Mar. 2025).
- [151] SADEGHI, A.-R., AND WACHSMANN, C. Security and privacy challenges in the age of artificial intelligence. *IEEE Security & Privacy* 18, 2 (2020), 15–26.
- [152] SADEGHI, A.-R., AND WACHSMANN, C. Security and privacy challenges in the age of artificial intelligence. *IEEE Security & Privacy* 18, 2 (2020), 15–26.
- [153] SALAH, A. R. M. Jailbreaking llms: An in-depth security analysis of prompt injection vulnerabilities.
- [154] SAMUDRALA, V. K., PULAKHANDAM, W., AND VALLU, V. R. Mitigating security risks in robotic telesurgery with securesurginet protocols for replay, brute force, and session hijacking attacks. *Advances in computational intelligence and robotics book series* (2025).
- [155] SANTIS, F. D., HUANG, K., VALENTIM, R., GIORDANO, D., MELIA, M., HOUIDI, Z. B., AND ROSSI, D. CFA-Bench: Cybersecurity forensic llm agent benchmark and testing.
- [156] SARKER, I. H. Cyber offensive ai: Threats, challenges, and research directions. *Advanced Science, Engineering and Medicine* 15, 10 (2023), 54–63.
- [157] SARKER, I. H., JANICKE, H., BUCZAK, A. L., AND GUVEN, E. Towards an ai-enhanced cyber threat intelligence processing pipeline. *Electronics* 13, 11 (2024), 2021.
- [158] SARKER, I. H., RAHMAN, M., ET AL. Autonomous ai agents for cyber attack process optimization. *International Journal of Computer Science and Network Security* 23, 4 (2023), 45–56.
- [159] SCHREIBER, A., AND SCHREIBER, I. Ai for cyber-security risk: harnessing ai for automatic generation of company-specific cybersecurity risk profiles. *Information & computer security* (2025).
- [160] SCHWARTZ, Y., BEN-SHIMOL, L., MIMRAN, D., ELOVICI, Y., AND SHABTAI, A. Llmcloudhunter: Harnessing llms for automated extraction of detection rules from cloud-based cti. In *Proceedings of the ACM on Web Conference 2025* (2025), pp. 1922–1941.
- [161] SHARIF, M. Practical inference-time attacks against machine-learning systems and a defense against them.
- [162] SHARMA, A., REDDY, M. U., LATHIGARA, A., AND VERMA, A. Automated malware classification using deep learning neural networks. *2023 IEEE International Conference on ICT in Business Industry Government (ICTBIG)* (Dec. 2023).
- [163] SHAZIYA, S. Artificial intelligence in cybersecurity: a decade of research trends and developments through bibliometric analysis. *International Journal of Advanced Research in Computer Science* (2025).
- [164] SHEN, Y., ET AL. Large language model alignment: A survey. *arXiv preprint arXiv:2309.15025* (2023).
- [165] SHETH, A., ACHANTA, A., MATAM, P., PATEL, A., SHARMA, P., JANAPAREDDY, N. V. P., PATIL, B., AND GUDUR, V. Ai driven self-healing cybersecurity systems with agentic ai for adaptive threat response and resilience. *2025 IEEE Cloud Summit* (2025).
- [166] SHEVLANE, T., FARQUHAR, S., GARFINKEL, B., ET AL. Model evaluation for extreme risks. *arXiv preprint arXiv:2305.15324* (2023).
- [167] SHOKRI, R., STRONATI, M., SONG, C., AND SHMATIKOV, V. Membership inference attacks against machine learning models. In *IEEE Symposium on Security and Privacy* (2017), pp. 3–18.

- [168] SIEGEL, N., ET AL. Reinforcement learning for adaptive malware evasion. *IEEE Transactions on Information Forensics and Security* 16 (2021), 3818–3830.
- [169] SINGH, D. K. Profit protection 2.0: The future of large language models (llms) in data security. *World Journal Of Advanced Research and Reviews* (2024).
- [170] SINGH, R., TARIQ, S., JALALVAND, F., CHHETRI, M. B., NEPAL, S., PARIS, C., AND LOCHNER, M. Llms in the soc: An empirical study of human-ai collaboration in security operations centres.
- [171] SPYROS, A., KORITSAS, I., PAPOUTSIS, A., PANAGIOTOU, P., CHATZAKOU, D., KAVALLIEROS, D., TSIKRIKA, T., VROCHIDIS, S., AND KOMPATSIARIS, I. Ai-based holistic framework for cyber threat intelligence management. *IEEE Access* 13 (2025), 20820–20846.
- [172] STEINHARDT, J., KOH, P. W., AND LIANG, P. Certified defenses for data poisoning attacks. In *Advances in Neural Information Processing Systems (NeurIPS)* (2017), pp. 3517–3529.
- [173] SUN, J., LI, A., WANG, B., YANG, H., LI, H., AND CHEN, Y. Soteria: Provable defense against privacy leakage in federated learning from representation perspective.
- [174] SUN, R., CHANG, J., PEARCE, H., XIAO, C., LI, B., WU, Q., NEPAL, S., AND XUE, M. SoK: Unifying cybersecurity and cybersafety of multimodal foundation models with an information theory approach. *arXiv preprint arXiv:2411.11195* (2024).
- [175] SURYOTRISONGKO, H., SHARMA, S., AND MODI, T. Ai-augmented soc: A survey of llms and agents for security operations automation. *Cybersecurity* 5, 4 (2025), 95.
- [176] SZEGEDY, C., ZAREMBA, W., SUTSKEVER, I., ET AL. Intriguing properties of neural networks. In *International Conference on Learning Representations (ICLR)* (2014).
- [177] TABASSI, E., ET AL. Artificial intelligence risk management framework (ai rmf 1.0). Tech. Rep. NIST AI 100-1, National Institute of Standards and Technology, Gaithersburg, MD, January 2023. [Online; accessed 5 January 2026].
- [178] TAKEMOTO, K. All in how you ask for it: Simple black-box method for jailbreak attacks. *ArXiv* (Jan. 2024).
- [179] TELLACHE, A., KORBA, A. A., MOKHTARI, A., MOLDOVAN, H., AND GHAMRI-DOUDANE, Y. Advancing autonomous incident response: Leveraging llms and cyber threat intelligence.
- [180] TIHANYI, N., FERRAG, M., JAIN, R., BISZTRAY, T., AND DEBBAB, M. Cybermetric: A benchmark dataset based on retrieval-augmented generation for evaluating llms in cybersecurity knowledge. *2024 IEEE International Conference on Cyber Security and Resilience (CSR)* (Feb. 2024).
- [181] TOMIČIĆ, I., GRD, P., AND BERNIK, A. Advancing the threat intelligence with ai: An overview, taxonomy and roadmap. *Journal of Universal Computer Science* (2025).
- [182] TRAMÈR, F., AND BONEH, D. Adversarial training and robustness against adaptive attacks. *Advances in Neural Information Processing Systems* (2020).
- [183] TURNER, A., TSIPRAS, D., AND MADRY, A. Clean-label backdoor attacks. In *Advances in Neural Information Processing Systems (NeurIPS)* (2019).
- [184] UCHEJI, C., EKENEME, J., AND EZEKWE, C. Global trends in ai-driven cybersecurity: A systematic and bibliometric analysis. *Asian Journal of Research in Computer Science* (2025).
- [185] VAISHALI, V. The evolution of agentic ai in cybersecurity: From single llm reasoners to multi-agent systems and autonomous pipelines. *arXiv (Cornell University)* (Dec. 2025).
- [186] VELDANDA, A., AND GARG, S. On evaluating neural network backdoor defenses. *ArXiv* (Oct. 2020).
- [187] VEPRYTSKA, O., KHARCHENKO, V., AND ILLIASHENKO, O. Cybersecurity and artificial intelligence: Triad-based analysis and attacks review. *Cybernetics and Information Technologies* (2025).
- [188] VERMA, A., KRISHNA, S., GEHRMANN, S., SESHADRI, M., PRADHAN, A., AULT, T., BARRETT, L., RABINOWITZ, D., DOUCETTE, J., AND PHAN, N. Operationalizing a threat model for red-teaming large language models (llms). *arXiv preprint arXiv:2407.14937* (2024).
- [189] WAIRAGADE, A. Strategic management of ai-powered cybersecurity systems: A systematic review. *Journal of Engineering Research and Reports* (2025).
- [190] WALKER, A., AND SENGUPTA, S. Malware family fingerprinting through behavioral analysis. *2020 IEEE International Conference on Intelligence and Security Informatics (ISI)* (Nov. 2020).
- [191] WANG, J., PAL, A., YANG, Q., KANT, K., ZHU, K., AND GUO, S. Collaborative machine learning: Schemes, robustness, and privacy. *IEEE Transactions on Neural Networks and Learning Systems* (May 2022).
- [192] WANG, L., WANG, J., JUNG, K., THIAGARAJAN, K., WEI, E., SHEN, X., CHEN, Y., AND LI, Z. From sands to mansions: Enabling automatic full-life-cycle cyberattack construction with llm. *arXiv.org* (2024).
- [193] WANG, Q., ZONG, B., LIN, Y., LI, Z., AND LUO, X. The application of big data and artificial intelligence technology in enterprise information security management and risk assessment. *Journal of Organizational and End User Computing (JOEUC)* 35, 1 (2023), 1–15.
- [194] WEIDINGER, L., ET AL. Ethical and social risks of harm from language models. *arXiv preprint arXiv:2112.04359* (2021).
- [195] WEN, C., CAI, Y., ZHANG, B., SU, J., XU, Z., LIU, D., QIN, S., MING, Z., AND CONG, T. Automatically inspecting thousands of static bug warnings with large language model: How far are we? *ACM Transactions on Knowledge Discovery from Data* (Mar. 2024).
- [196] WILHELM, C., AND YOUNIS, A. A. A threat analysis methodology for security requirements elicitation in machine learning based systems. *2020 IEEE 20th International Conference on Software Quality, Reliability and Security Companion (QRS-C)* (Nov. 2020).
- [197] WILLIAMS, D., CLARK, C., MCGAHAN, R., POTTEIGER, B., COHEN, D., AND MUSAU, P. Discovery of ai/ml supply chain vulnerabilities within automotive cyber-physical systems. *2022 IEEE International Conference on Assured Autonomy (ICAA)* (Feb. 2022).
- [198] WINSTON, C., AND JUST, R. A taxonomy of failures in tool-augmented llms. *2025 IEEE/ACM International Conference on Automation of Software Test (AST)* (Apr. 2025).
- [199] XIA, C., PALTENGHI, M., TIAN, J. L., PRADEL, M., AND ZHANG, L. Fuzz4all: Universal fuzzing with large language models. *2024 IEEE/ACM 46th International Conference on Software Engineering (ICSE)* (Aug. 2023).
- [200] XIAO, N., LANG, B., WANG, T., AND CHEN, Y. Apt-mm: An advanced persistent threat actor attribution method based on multimodal and multilevel feature fusion. *ArXiv* (Feb. 2024).
- [201] XU, H., WANG, S., LI, N., WANG, K., ZHAO, Y., CHEN, K., YU, T., LIU, Y., AND WANG, H. Large language models for cyber security: A systematic literature review. *ACM Transactions on Software Engineering and Methodology* (2025).
- [202] XU, J., NING, P., ZHOU, C., ET AL. Cybersecurity knowledge graphs. *Knowledge and Information Systems* 65 (2023), 3511–3531.
- [203] XU, X., KONG, K., LIU, N., CUI, L.-Z., WANG, D., ZHANG, J., AND KANKANHALLI, M. S. An llm can fool itself: A prompt-based adversarial attack. *ArXiv* (Oct. 2023).
- [204] YAMAMOTO, T., SATO, M., AND TANAKA, K. Comparative investigation of traditional machine-learning models and transformer models for phishing email detection. *Electronics* 13, 24 (2024), 4877.
- [205] YAO, Y., DUAN, J., XU, K., CAI, Y., SUN, Z., AND ZHANG, Y. A survey on large language model (llm) security and privacy: The good, the bad, and the ugly. *High-Confidence Computing* (Feb. 2024).
- [206] ZHANG, A. K., PERRY, N., DULEPET, R., JONES, E., LIN, J. W., JI, J. F., MENDERS, C., HUSSEIN, G., LIU, S., JASPER, D., PEE-TATHAWATCHAI, P., GLENN, A., SIVASHANKAR, V., ZAMOSHCHIN, D., GLIKBARG, L., ASKARYAR, D., YANG, M., ZHANG, T., AL-LURI, R., TRAN, N., SANGPISIT, R., YIORKADJIS, P., OSELE, K., RAGHUPATHI, G., BONEH, D., HO, D. E., AND LIANG, P. Cybench: A framework for evaluating cybersecurity capabilities and risk of language models. *arXiv.org* (2024).
- [207] ZHANG, C., LIU, H., ZENG, J., YANG, K., LI, Y., AND LI, H. Prompt-enhanced software vulnerability detection using chatgpt. *arXiv preprint arXiv:2308.12697* (2024).
- [208] ZHANG, J., WEN, H., DENG, L., XIN, M., LI, Z., LI, L., ZHU, H., AND SUN, L. Hackmentor: Fine-tuning large language models for cybersecurity. *2023 IEEE 22nd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)* (Oct. 2023).
- [209] ZHANG, Z., TAO, G., SHEN, G., AN, S., XU, Q., LIU, Y., YE, Y., WU, Y., AND ZHANG, X. Pelican: Exploiting backdoors of naturally trained deep learning models in binary code analysis.
- [210] ZHAO, J., ET AL. Ai assisted network reconnaissance and profiling. *Journal of Cybersecurity* 2, 1 (2021), 112–125.
- [211] ZHU, L., LIU, Z., AND HAN, S. Deep leakage from gradients. In *Advances in Neural Information Processing Systems* (2019).